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// Online C compiler to run C program online
#include <stdio.h>

int linearSearch(int arr[], int size, int target){
    for(int i = 0; i<size; i++){
        if( arr[i] == target) {
            return i;
        }
    }
    return -1;
}

int binarySearch(int arr[], int size, int target){
    int low = 0, high = size - 1;

    while(low <= high){
        int mid = (low + high) / 2;

        if(arr[mid] == target){
            return mid;
        }
        else if(arr[mid] < target){
            low = mid + 1;
        }
        else{
            high = mid - 1;
        }
    }
    return -1;
}

int main() {
    int size, target;

    printf("Enter your array size: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter your array value (must be shorted)\n");
    for(int i = 0; i<size; i++){
        printf("%d index value: ", i);
        scanf("%d", &arr[i]);
    }

    printf("Array is: ");
    for(int i = 0; i<size; i++){
        printf("%d ", arr[i]);
    }
    printf("\n");

    printf("Enter your search element: ");
    scanf("%d", &target);
```

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// int rslt = linearSearch(arr, size, target);

int rslt = binarySearch(arr, size, target);

if (rslt != -1) {
    printf("Element %d found at index %d\n", target, rslt);
}
else {
    printf("Element %d not found in the array\n", target);
}
}
```